MP2520F Development Kit4 GPIO Define Ver 4.1.1

MDMP2520FDEB411GPIO

2005.6.2

No	GPIO	Pin Name	Net Name	In/Out	GPIO/ALT	Description	USED	Pull Up/Pull Down	TP
1	A[15..0]	GPIOA[15]/VD[23]	GPIOA_15	OUT	G			External P/U	
2		GPIOA[14]/VD[22]	VD22		A1	GPIOA[9..0] : LCD GPIOA[3..0] : ENCODER GPIOA[14..3, 1..0] : Reset Configuration setting		External P/U	
3		GPIOA[13]/VD[21]	VD21		A1			External P/D	
4		GPIOA[12]/VD[20]	VD20		A1			External P/D	
5		GPIOA[11]/VD[19]	VD19		A1			External P/U	
6		GPIOA[10]/VD[18]	VD18		A1			External P/U	
7		GPIOA[09]/VD[17]	VD17		A1			External P/D	
8		GPIOA[08]/VD[16]	VD16		A1			External P/U	
9		GPIOA[07]/VD[15]	VD15		A1			External P/U	
10		GPIOA[06]/VD[14]	VD14		A1			External P/D	
11		GPIOA[05]/VD[13]	VD13		A1			External P/U	
12		GPIOA[04]/VD[12]	VD12		A1			External P/D	
13		GPIOA[03]/VD[11]	VD11		A1			External P/U	
14		GPIOA[02]/VD[10]	VD10		A1				
15		GPIOA[01]/VD[09]	VD9		A1			External P/D	
16		GPIOA[00]/VD[08]	VD8		A1			External P/D	
17	B[15..0]	GPIOB[15]/VD[07]	VD7		A1	GPIOB[15..8] : LCD, ENCODER GPIOB[15..13] : Reset Configuration setting		External P/D	
18		GPIOB[14]/VD[06]	VD6		A1			External P/D	
19		GPIOB[13]/VD[05]	VD5		A1			External P/D	
20		GPIOB[12]/VD[04]	VD4		A1				
21		GPIOB[11]/VD[03]	VD3		A1				
22		GPIOB[10]/VD[02]	VD2		A1				
23		GPIOB[09]/VD[01]	VD1		A1				
24		GPIOB[08]/VD[00]	VD0		A1				
25		GPIOB[07]/CLKH	CLKHO		A1				3
26		GPIOB[06]/DE	ACTIVE		A1				
27		GPIOB[05]/HSYNC	HSYNC		A1				
28		GPIOB[04]/VSYNC	VSYNC		A1				
29		GPIOB[03]/POL	nCX_RST	OUT	G	nRESET signal for CX25874 - Active LOW		External P/U	

30		GPIOB[02]/CLKV	CX_PD	OUT	G	Power Down signal for CX25874 - Active HIGH		External P/D	
31		GPIOB[01]/PS	GPIO_SCL	OUT	G	GPIO I2C clock signal		External P/U	
32		GPIOB[00]/FG	GPIO_SDA	IN/OUT	G	GPIO I2C data signal		External P/U	
33		GPIOC[15]/ID[15]	KEY_UP	IN	G	SW3		External P/U	
34		GPIOC[14]/ID[14]	KEY_DOWN	IN	G	SW4		External P/U	
35		GPIOC[13]/ID[13]	KEY_MENU	IN	G	SW5		External P/U	
36		GPIOC[12]/ID[12]	KEY_EXIT	IN	G	SW6		External P/U	
37		GPIOC[11]/ID[11]	KEY_ENTER	IN	G	SW7		External P/U	
38		GPIOC[10]/ID[10]	CIS_PW	OUT	G	Power enable test signal of U26(VCC2P5)		External P/U	65
39		GPIOC[09]/ID[09]	DU_PW	OUT	G	Power enable test signal of U5(VCC3P3_DU)		External P/U	61
40	C[15..0]	GPIOC[08]/ID[08]	EA_PW	OUT	G	Power enable test signal of U17(VCC3P3_EA)		External P/U	59
41		GPIOC[07]/ID[07]	ID7		A1				
42		GPIOC[06]/ID[06]	ID6		A1				
43		GPIOC[05]/ID[05]	ID5		A1				
44		GPIOC[04]/ID[04]	ID4		A1	ID[7..0] : CAMERA Input video data sianal			
45		GPIOC[03]/ID[03]	ID3		A1				
46		GPIOC[02]/ID[02]	ID2		A1				
47		GPIOC[01]/ID[01]	ID1		A1				
48		GPIOC[00]/ID[00]	ID0		A1				
49		GPIOD[13]/TX[3]	UART3_Tx		A1	UART channel 3 Tx signal (Connected to SOCKET_1)			
50		GPIOD[12]/RX[3]/SSPEXTCLKG	UART3_Rx		A1	UART channel 3 Rx signal (Connected to SOCKET_1)			
51		GPIOD[11]/TX[2]	UART2_Tx		A1	UART channel 2 Tx signal (Connected to SOCKET_1)			
52		GPIOD[10]/RX[2]	UART2_Rx		A1	UART channel 2 Rx signal (Connected to SOCKET_1)			
53		GPIOD[09]/RX[1]	CF1_nRST	OUT	G	RESET signal for CF card - default : ACTIVE High			
54		GPIOD[08]/TX[1]	CF1_RDYnBSY1	IN	G	Ready & Busy signal of CF		External P/U	
55		GPIOD[07]/NRTS	CF1_S1_VS1	IN	G	Voltage Sense 1 signal of CF		External P/U	
56		GPIOD[06]/NCTS	CF1_S1_VS2	IN	G	Voltage Sense 2 signal of CF		External P/U	
57	D[13..0]	GPIOD[05]/NDTR	nCF1_CD	IN	G	Card Detect signal of CF - ACTIVE Low		External P/U	

58	D[15..0]	GPIOD[04]/NDSR	nUSB_RST	OUT	G	nRESET signal for NET2272(USB2.0 Device) - ACTIVE Low		External P/U	
59		GPIOD[03]/NDCD	USBD_CON	OUT	G	Control USB1.1 Device D+ Siganl to Pull-up(or Pull-down)			54
60		GPIOD[02]/NRIO	SD_WP	IN	G	Write Protect signal of SD		External P/U	
61		GPIOD[01]/RX0	RX0		A1	UART channel 0 Rx signal - Connected to SOCKET_3 via MAX3232 RS-232 transceiver			
62		GPIOD[00]/TX0	TX0		A1	UART channel 0 Tx signal - Connected to SOCKET_3 via MAX3232 RS-232 transceiver			
63		GPIOE[15]/ZD[15]	ZD15		A1				
64		GPIOE[14]/ZD[14]	ZD14		A1				
65		GPIOE[13]/ZD[13]	ZD13		A1				
66		GPIOE[12]/ZD[12]	ZD12		A1				
67		GPIOE[11]/ZD[11]	ZD11		A1				
68		GPIOE[10]/ZD[10]	ZD10		A1				
69		GPIOE[09]/ZD[09]	ZD9		A1				
70	E[15..0]	GPIO[08]/ZD[08]	ZD8		A1	ZD[15..0] : IDE, CF, USB2.0, Ethernet Controller ZD[7..0] : NAND FLASH			
71		GPIO[07]/ZD[07]	ZD7		A1			External P/D	
72		GPIO[06]/ZD[06]	ZD6		A1				
73		GPIO[05]/ZD[05]	ZD5		A1				
74		GPIO[04]/ZD[04]	ZD4		A1				
75		GPIO[03]/ZD[03]	ZD3		A1				
76		GPIO[02]/ZD[02]	ZD2		A1				
77		GPIO[01]/ZD[01]	ZD1		A1				
78		GPIO[00]/ZD[00]	ZD0		A1				
79		GPIOF[09]/ZA[25]/OM[3]	OM3		A1	Not used	NO	External P/D	
80		GPIOF[08]/ZA[24]/OM[2]	OM2		A1	Not used	NO	External P/D	
81		GPIOF[07]/ZA[23]/OM[1]	OM1		A1	Not used	NO	External P/D	
82		GPIOF[06]/ZA[22]	TSPX_EN	OUT	G	FET Control signal for Touch Panel - X Plus			
83		GPIOF[05]/ZA[21]	TSMX_EN	OUT	G	FET Control signal for Touch Panel - X Minus			

84	F[9..0]	GPIOF[04]/ZA[20]	TSPY_EN	OUT	G	FET Control signal for Touch Panel - Y Plus			
85		GPIOF[03]/ZA[19]	TSMY_EN	OUT	G	FET Control signal for Touch Panel - Y Minus			
86		GPIOF[02]/ZA[18]	GPIOF_2	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		15
87		GPIOF[01]/ZA[17]	GPIO_P0EN	OUT	G	Control signal for CF Power Control IC - Connected to U18			
88		GPIOF[00]/ZA[16]	GPIO_P1EN	OUT	G	Control signal for CF Power Control IC - Connected to U18			
89	G[15..0]	GPIOG[15]/ZA[15]	DEC_CE	OUT	G	Chip enable signal of Video DECODER (SAA7113) - Low : power sleep mode - High : normal mode		External P/U	36
90		GPIOG[14]/ZA[14]	GPIOG_14	IN	G	Connected to SW1 (Dip SW)		External P/U	38
91		GPIOG[13]/ZA[13]	GPIOG_13	IN	G			External P/U	39
92		GPIOG[12]/ZA[12]	GPIOG_12	IN	G			External P/U	40
93		GPIOG[11]/ZA[11]	GPIOG_11	IN	G			External P/U	41
94		GPIOG[10]/ZA[10]	ZA10		A1	ZA[10..0] MCU-C bank Adrress Bus			
95		GPIOG[09]/ZA[09]	ZA9		A1				
96		GPIOG[08]/ZA[08]	ZA8		A1				
97		GPIOG[07]/ZA[07]	ZA7		A1				
98		GPIOG[06]/ZA[06]	ZA6		A1				
99		GPIOG[05]/ZA[05]	ZA5		A1				
100		GPIOG[04]/ZA[04]	ZA4		A1				
101		GPIOG[03]/ZA[03]	ZA3		A1				
102		GPIOG[02]/ZA[02]	ZA2		A1				
103		GPIOG[01]/ZA[01]	ZA1		A1				
104		GPIOG[00]/ZA[00]/DQMZ[0]/BTENB[0]	ZA0		A1				
105		GPIOH[06]/XDOFF	USB_DETECT	IN	G	Detect signal for USB1.1 Host port - Low : Status that PC is not connected - High : Status that PC is connected			
106		GPIOH[05]/VCLKIN	CX_CLKI		A1	Clock input signal from Video ENCODER (CX25874)			

107	H[6..0]	GPIOH[04]/ICLKIN	CIS_CLKI		A1	Clock input signal from CMOS Camera module			
108		GPIOH[03]/IHSYNC	CIS_HSYNC		A1	HSYNC signal from CMOS Camera module			
109		GPIOH[02]/IVSYNC	CIS_VSYNC		A1	VSYNC signal from CMOS Camera module			
110		GPIOH[01]/ICLKOUT	CIS_CLKO		A1	Clock output signal to CMOS Camera module			
111		GPIOH[00]/EXTCLKO	GPIOH_0	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		
112		GPIOI[15]/CBBUSREG	GPIOI_15	IN	G	N.C. pin - Can not use any purpose	NO	External P/D	43
113		GPIOI[14]/CBBUSGNT	nSD_CD	IN	G	Card detect signal of SD/MMC		External P/U	42
114		GPIOI[13]/NPWAIT	nWAIT		A1	wait signal of IDE, CF, and Ethernet controller			
115		GPIOI[12]/DQMz01/BTENB[1]	ETHER_nBTENB1		A1	ETHERNET BIT ENB		External P/U	23
116		GPIOI[11]/NSCS[03]/656IN[07]	LCD_PW	OUT	G	Power enable test signal of U33(VCC3P3_LCD)		External P/U	44
117		GPIOI[10]/NSCS[02]/656IN[06]	USB_nSCS2		A1	Chip Select signal for USB2.0(NET2272)			20
118		GPIOI[09]/NSCS[01]/656IN[05]	ETHER_nSCS1		A1	Chip Select signal for Ethernet(CS8900)			19
119		GPIOI[08]/NSCS[00]	nSCS0		A1	Chip Select signal for IDE		External P/U	18
120		GPIOI[07]/NPCS[01]	CF1_nPCE2		A1	High byte enable signal for CF		External P/U	
121	I[15..0]	GPIOI[06]/NPCS[00]	CF1_nPCE1		A1	Low byte enable signal for CF		External P/U	
122		GPIOI[05]/NICS[01]	HARD_nICS1	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO	External P/U	
123		GPIOI[04]/NICS[00]	HARD_nICS0	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO	External P/U	
124		GPIOI[03]/NNFCE[03]/656IN[04]	GPIOI_3	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		
125		GPIOI[02]/NNFCE[02]/656IN[03]	nNFWP	OUT	G	Write Protect signal for NAND Flash - ACTIVE Low		External P/U	12
126		GPIOI[01]/NNFCE[01]/656IN[02]	GPIOI_1	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		

127		GPIOI[00]/NNFCE[00]	nNFCS0		A1	Chip Select signal for NAND Flash		External P/U	11
128		GPIOJ[15]/CBREQ	GPIOJ_15	IN	G	N.C. pin - Can not use any purpose	NO	External P/D	
129		GPIOJ[14]/CBACK	USB_nIRQ	IN	G	IRQ signal of USB2.0 (NET2272)		External P/U	
130		GPIOJ[13]/CBRDY	USB_PO_EN	OUT	G	Power Enable signal for USB Host		External P/U	
131		GPIOJ[12]/NPOE	nPOE		A1	Read Enable signal for CF/USB2.0/IDE/Ethernet			
132		GPIOJ[11]/NPWE/WRITE_CRB	nPWE		A1	Write Enable signal for CF/USB2.0/IDE/Ethernet			
133		GPIOJ[10]/NPOIR	nPIOR		A1	IO Read Enable signal for CF, Command Latch Enable signal for NAND			6
134	J[15..0]	GPIOJ[09]/NPIOW	nPIOW		A1	IO Write Enable signal for CF, Address Latch Enable signal for NAND			7
135		GPIOJ[08]/PSKSEL	nPSKTSEL		A1	Socket Select signal for CF			
136		GPIOJ[07]/NPREG	CF1_NPREG		A1	Register Select signal for CF			
137		GPIOJ[06]/NPIOS16	nPIOIS16		A1	IOIS16 signal of CF		External P/U	
138		GPIOJ[05]/RDNWR	CF1_RDnWR		A1	Buffer direction control signal for CF			
139		GPIOJ[04]/NBUFOE	nBUFOE		A1	Buffer Output Enable signal - Not used	NO		21
140		GPIOJ[03]/NBUFENB	nBUFENB		A1	Buffer Enable signal for CF			22
141		GPIOJ[02]/RNB	NFRnB		A1	Ready nBusy signal of NAND			10
142		GPIOJ[01]/NNFOE	nNFOE		A1	Read Enable signal for NAND			8
143		GPIOJ[00]/NNNFWE	nNFWE		A1	Write Enable signal for NAND			9
144		GPIOK[07]/DERQ3/STVD	LCD_RL	IN/OUT	G	Not used	NO	External P/D	
145		GPIOK[06]/DREQ2/656IN[01]	LCD_UD	IN/OUT	G	Not used	NO	External P/D	
146		GPIOK[05]/DREQ[01]	USB_DREQ1		A1	DMA Request signal of USB2.0 Controller - meaningless		External P/D	
147		GPIOK[04]/DREQ[00]	IDE_DMAREQ		A1	DMA Request signal of IDE - meaningless	NO		
148	K[7..0]	GPIOK[03]/DVAL[03]/STHR	GPIOK_3	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		
149		GPIOK[02]/DVAL[02]/656IN[00]	GPIOK_2	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		

150		GPIOK[01]/DVAL[01]	USB_DACK1		A1	DMA Acknowledge signal for USB2.0 Controller - meaningless			
151		GPIOK[00]/DVAL[00]	IDE_DMAACK		A1	DMA Acknowledge signal for IDE - meaningless	NO		
152		GPIOL[14]/PWMOUT[03]/SSPIN	CF1_FAIL	IN	G	Combination signal of BVD1 & BVD2 of CF - Not used now…			
153		GPIOL[13]/PWMOUT[02]/PPM	IR_REMO		A2	Remocon Input signal		External P/U	
154		GPIOL[12]/PWMOUT[01]	LCD_BR_CON		A1	Brightness control signal for LCD Backlight			
155		GPIOL[11]/PWMOUT[00]	GPIOL_11	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO	External P/D	
156		GPIOL[10]/ABITCLK/I2SCLK	AC97_BITCLK / I2S_BITCLK		A1 / A2	Bit Clock signal of AC97 / Bit Clock signal for I2S			
157		GPIOL[09]/ADATAIN/I2SDATAIN	AC97_DATIN / I2S_DATIN		A1 / A2	Data Input from AC97 / Data Input from I2S			
158	L[14..0]	GPIOL[08]/ASYNC/I2SSYNC	AC97_SYNC / I2S_SYNC		A1 / A2	Sync signal for AC97 / Sync signal for I2S			
159		GPIOL[07]/NARST/I2SACLK	AC97_nRST / I2S_nRST		A1 / A2	Reset signal for AC97 - ACTIVE Low / Reset signal for I2S - ACTIVE Low			
160		GPIOL[06]/ADATAOUT/I2SDATAOUT	AC97_DATOUT / I2S_DATOUT		A1 / A2	Data Output for AC97 / Data Output for I2S			
161		GPIOL[05]/SDICLK	SD_CLK		A1	Clock signal for SD/MMC			
162		GPIOL[04]/SDICMD	SD_CMD		A1	SD/MMC Command signal		External P/U	
163		GPIOL[03]/SDIDAT[03]/MSBS	SD_DAT3		A1			External P/U	
164		GPIOL[02]/SDIDAT[02]/MSDAT	SD_DAT2		A1	SD/MMC Data		External P/U	
165		GPIOL[01]/SDIDAT[01]	SD_DAT1		A1			External P/U	
166		GPIOL[00]/SDIDAT[00]	SD_DAT0		A1			External P/U	
167		GPIOM[08]/VD656[07]/TSISYNC/LRCK	V656_D7		A1				
168		GPIOM[07]/VD656[06]/TSIDP/BCK	V656_D6		A1				
169		GPIOM[06]/VD656[05]/STIERR/SDAT	V656_D5		A1				
170		GPIOM[05]/VD656[04]/CDRX	V656_D4		A1	V656[7..0] Video Input signal of Video DECODER(SAA7113)			
171	M[8..0]	GPIOM[04]/VD656[03]/CDTX	V656_D3		A1				
172		GPIOM[03]/VD656[02]/TSDI[00]	V656_D2		A1				
173		GPIOM[02]/VD656[01]	V656_D1		A1				

174		GPIOM[01]/VD656[00]	V656_D0		A1				
175		GPIOM[00]/VDCLKIN/TSICLK	V656_CLK		A1	Video Clock Input signal of Video DECODER(SAA7113)			
176	N[7..0]	GPION[07]/TSIDI[07]/TSISYNC(CDRX)	nIDE_RST	OUT	G	Reset signal for IDE - ACTIVE Low		External P/U	
177		GPION[06]/TSIDI[06]/TSIDP(CDTX)	VS_1	OUT	G	Video Output select VS_1 VS_2 OUT 0 X YPbPr 1 0 CVBS,S-VIDEO 1 1 RGB			
178		GPION[05]/TSIDI[05]/TSIERR(LRCK)	VS_2	OUT	G				
179		GPION[04]/TSIDI[04]/TSIDI[00](BCK)	ETHER_IRQ	IN	G	IRQ signal from Ethernet controller		External P/D	
180		GPION[03]/TSIDI[03]/TSICLK(SDAT)	HARD_IRQ	IN	G	IRQ signal from IDE		External P/D	
181		GPION[02]/TSIDI[02]/MSINS	GPION_2	IN/OUT	G	Connected to CON5 - Can be used as GPIO Input or Output	NO		55
182		GPION[01]/TSIDI[01]/MSCLK	SLEEP	OUT	G	LED(D25) ON/OFF control signal - High : LED ON (indicate Normal mode) - Low : LED OFF (indicate Sleep mode)			56
183		GPION[00]/TSIDI[00]	BL_ON_OFF	OUT	G	LCD backlight ON/OFF signal		External P/D	2
184	O[5..0]	GPIOO[05]/SPDIFIN	GPIO_P0VCC5EN	OUT	G	Control signal for CF Power Control IC - Connected to U18			
185		GPIOO[04]/SPDIFOUT	GPIO_P0VCC3EN	OUT	G	Control signal for CF Power Control IC - Connected to U18			
186		GPIOO[03]/SSPCLK	CIS_RESET	OUT	G	Reset signal for Camera module			
187		GPIOO[02]/SSPFRM	CIS_PWDN	OUT	G	Power down signal for Camera module			35
188		GPIOO[01]/SSPDAT	GPIOO_1	OUT	G	Bus switch selection signal related to NAND Flash - High : Large Block NAND boot(PRE mode) case - Low : Small Block NAND boot case (default)		External P/D	52
189		GPIOO[00]/1WIRE	GPIOO_0	OUT	G	N.C. pin	NO		53